DARSHAN PATIL

Senior Generative AI Engineer | LLMs, RAG, Agentic AI & Multi-Agent Systems
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PROFESSIONAL SUMMARY

6.4 years experienced AI Engineer in software development firm with profound background in AI, specialising in designing and deploying NLP and LLM-based applications across HealthCare, Insurance, Enterprise domain and in EdTech. Proficient in leveraging Claude Code for AI-assisted software development, code generation, refactoring, test automation, and implementing Spec-Driven Development (SDD) practices to transform business requirements into executable specifications, ensuring traceability, quality, and faster delivery throughout the AI Development Lifecycle (ADLC). I have expertise in deploying RAG systems, Prompt Engineering and creating production ready AI assistants using OpenAI, Azure AI, LangChain and Vector Databases also managing all aspects of Artificial Intelligence Project from collecting data, analyzing raw data, EDA, Statistics, model building, deploying till source code management. Brings excellent communication and ability to solve real-world business problems with hands-on experience in building Machine Learning, NLP, Deep Learning and LLM's.

TECHNICAL SKILLS

AI Frameworks	LangChain, LangGraph, CrewAI, AutoGen, Guardrails, MCP, Hugging Face Transformers, LlamaIndex, PEFT, LoRA, Hugging Face, AI Agent Workflows
RAG & Vector DBs	FAISS, Pinecone, pgvector, ChromaDB, Qdrant, Reranking, Hybrid Search, RAG, Weaviate, Semantic Search, Embeddings, Metadata-aware Chunking
LLM Evaluation	Ragas, Langfuse, Opik, TruLens
AI / ML	Machine Learning, NLP, Transformers, Generative AI, LLMs, RAG, Agentic AI, Multi-Agent Systems, Fine-tuning, Evaluation
AI Dev Tools	Claude Code, Cursor AI, GitHub Copilot, Amazon Q, Kiro, Roo Code, Antigravity, GitHub Spec Kit, Spec-Driven Development, AI-assisted Code Generation, Code Refactoring
Prompt Engineering	Chain-of-Thought, Few-Shot, ReAct, Prompt Templates, System Prompts, Prompt Engineering
Python Stack	Python 3.x, Python, FastAPI, Pandas, NumPy, Scikit-learn, Pytest, Streamlit
Cloud / MLOps	AWS, AWS EC2, S3, ECS, ECR, Fargate, Bedrock, SageMaker, Azure AI Foundry, Azure OpenAI, Azure Functions, Azure App Service, Blob, Databricks, MLflow, DVC, Docker
Machine Learning	Classification, Regression, Clustering, Predictive Maintenance, NLP, Statistics, EDA
Deep Learning	ANN, CNN, RNN, LSTM, GRU, BERT, Transformers, PyTorch, TensorFlow
Database	PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, SQL, DynamoDB
Dev Tools	GitHub, Git, GitHub Actions, Swagger, REST APIs, Postman

PROFESSIONAL EXPERIENCE

Senior AI Engineer @Nitor Infotech As an Ascendion — June 2025 – September 2026

WellSky — Project — November 2025 - March 2026

RPGLE to C#/.NET Migration using Agentic AI

Business Problem: Legacy IBM AS/400 RPGLE applications required modernisation into C#/.NET while preserving business rules, application behaviour and functional equivalence. Manual analysis and rewriting of large legacy codebases was time-consuming and error-prone.

Tech Stack: Agentic AI, Claude Code, Spec-Driven Development (SDD), RAG, LangChain, LLMs, Python, C#/.NET

- Worked on designing and implementing AI-powered solutions to automate migration of IBM AS/400 RPGLE applications into modern C#/.NET applications.
- Developed LLM-powered workflows for legacy code analysis, business rule extraction, code generation, validation and documentation.
- Designed Agentic AI workflows using specialised agents for source-code analysis, business-rule identification, C# code generation and migration validation.
- Applied RAG-based workflows to provide migration agents with relevant legacy application context, business rules, database structures and supporting documentation.
- Leveraged Claude Code for AI-assisted code generation, refactoring, debugging and test development.
- Applied Spec-Driven Development (SDD) to convert business and migration requirements into structured specifications for AI-assisted implementation.
- Designed prompt-engineering strategies to improve the accuracy and consistency of generated C#/.NET code.
- Worked with legacy RPGLE source code and database dependencies to maintain functional equivalence during modernisation.
- Collaborated with legacy-system SMEs and .NET teams to validate business rules and ensure successful modernisation.

Senior Software Engineer

WellSky — Project — July 2025 - November 2025

AI-Powered Clinical Documentation Assistant for Long-Term Care (LTC)

Business Problem: Clinical teams in Long-Term Care and Skilled Nursing Facility workflows spent significant time manually documenting patient-provider conversations. The solution automated transcription and structured nursing documentation.

Tech Stack: Python, Whisper, FastAPI, WebSockets, GCP, Firestore, AI Information Extraction

- Developed an AI-powered clinical documentation solution for Long-Term Care (LTC) and Skilled Nursing Facility (SNF) workflows.
- Built an end-to-end solution using Whisper, FastAPI and WebSockets to transcribe patient-provider conversations in real time.
- Designed FastAPI micro-services and WebSocket streaming infrastructure for live audio processing and low-latency transcription.

- Developed AI-driven information extraction workflows to automatically populate structured nursing assessment forms from unstructured clinical conversations.
- Designed Firestore-based storage for patient records, transcription history, assessment forms and audit tracking.
- Deployed and maintained cloud-native backend services on GCP with focus on scalability, reliability and secure healthcare data processing.
- Collaborated with clinical stakeholders to optimise documentation workflows and reduce manual charting effort.

Senior Data Scientist @Msys Technologies — August 2022 – May 2024

KEC International — Project — July 2023- May 2024

Tender Intelligence Chatbot for Electrical Transmission Line Projects

Business Problem: EPC, engineering and pre-bid teams needed to review large tender documents and quickly find technical, commercial and eligibility information required for bid decisions.

Tech Stack: Python, FastAPI, LangChain, Azure OpenAI, Weaviate, Embeddings, Azure App Service, Blob Storage, Azure Functions, ACR

- Developed a domain-specific chatbot to assist EPC, engineering and pre-bid teams in extracting actionable insights from electrical transmission-line tender documents.
- Leveraged RAG architecture using LangChain, Weaviate Vector Database and Azure OpenAI embeddings to enable context-aware question answering across multi-section tender PDFs.
- Processed tender sections including scope, BOQ, technical specifications, eligibility and commercial requirements.
- Developed a robust PDF parser and metadata-aware chunking pipeline to improve retrieval of clauses related to conductor specifications, tower types and financial qualifications.
- Used GPT-3.5 Turbo for final answer and summary generation and applied prompt-engineering techniques for response augmentation.
- Created a FastAPI-based application with authentication, feedback capture, chat logging and role-based access.
- Deployed the chatbot using Azure App Service, Blob Storage, Azure Functions, Azure OpenAI and Azure Container Registry.
- Reduced manual tender-review time by more than 70%, enabling project managers and pre-bid analysts to make faster go/no-go decisions.

Senior Data Scientist (RAG pipeline with frontend application)

Zuno General Insurance — Project — December 2022 - June 2023

Insurance Q&A Chatbot with RAG

Business Problem: Insurance users needed an easy way to ask questions about policy PDFs and receive answers grounded in their policy content rather than generic information.

Tech Stack: LangChain, Azure OpenAI, OpenAI Embeddings, Weaviate, FastAPI, Azure Functions, Blob Storage, Docker

- Developed an insurance-domain chatbot allowing users to interact with and ask questions about their policy PDFs.
- Designed and implemented an end-to-end RAG pipeline covering document extraction, chunking, embeddings, semantic retrieval and LLM-based response generation.
- Integrated Azure OpenAI and embedding models to deliver accurate, context-aware answers grounded in policy documents.
- Implemented Weaviate Vector Database for storing and retrieving policy-document embeddings.
- Used LangChain to orchestrate document processing, retrieval and LLM response generation.
- Developed FastAPI-based backend services for real-time chatbot interactions.
- Deployed cloud-based components for document storage and processing and collaborated with product and compliance stakeholders to improve accuracy and usability.
- Recognised for effective communication and collaboration with cross-functional teams.

Data Scientist @Alpha Information Technology — June 2019 – July 2022

Yash Technologies — Project — June 2021 - July 2022

Customer Churn Analysis

Business Problem: The business needed to identify customers likely to churn, segment them by risk and use model predictions to support targeted customer-retention strategies.

Tech Stack: Python, Scikit-learn, Machine Learning, FastAPI, Docker, AWS ECS/Fargate, S3, DynamoDB, NGINX

- Investigated customer churn using machine learning techniques to identify customers likely to leave.
- Collected customer data from databases, converted it into analysis-ready datasets and performed data cleaning, EDA and statistical analysis.
- Identified customer characteristics and behavioural patterns associated with churn.
- Categorised customers into three segments and predicted the segment in which each customer was likely to fall.
- Used model predictions to support business strategies for customers at risk of churn.
- Performed model development, evaluation and deployment using Python and Scikit-learn.
- Containerised the solution using Docker and deployed it using AWS ECS/Fargate.
- Used Amazon S3 for model storage and DynamoDB for customer data.
- Developed FastAPI APIs and used NGINX to route requests to application services.
- Used ECS task definitions and load-balancing capabilities to support scalable deployment.

CERTIFICATIONS AND RECOGNITIONS

- Got appreciation from clients for bringing value through AI/ML solutions and contributing to business outcomes.
- Recognised by project leads for effective delivery, communication and cross-functional collaboration.
- Deliver on promise and earn client trust every day.

EDUCATIONAL BACKGROUND

- B.Tech Aerospace Engineering from IIAEIT Pune (2010-2015)
- Higher Secondary College (2008-2010)